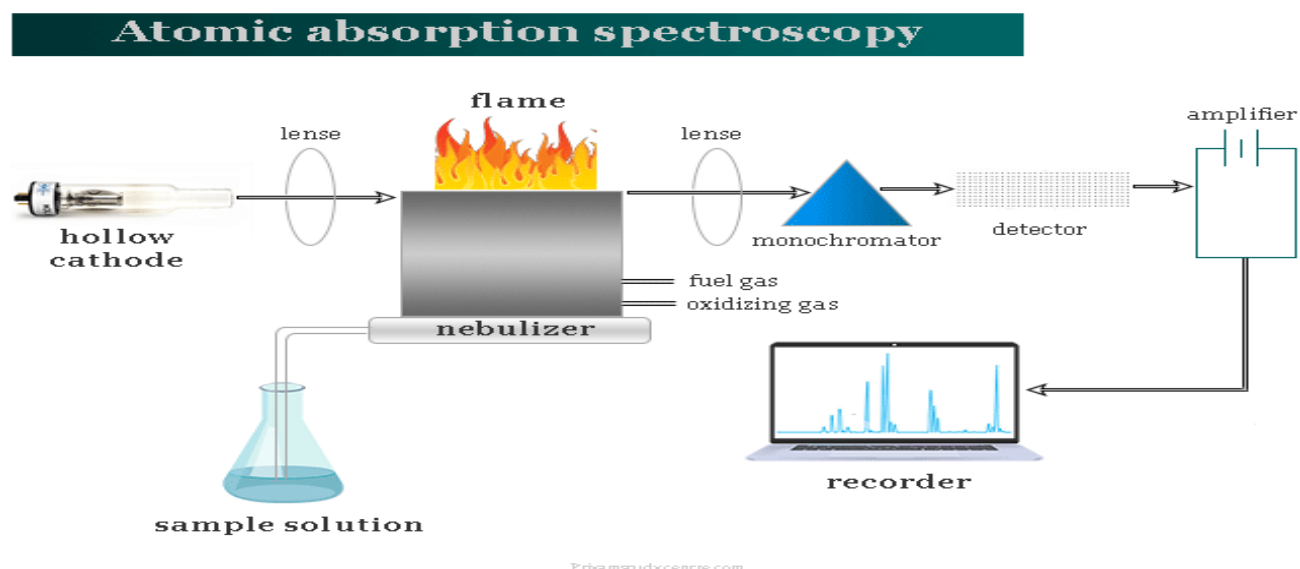


Atomic Absorption Spectroscopy and Green Chemistry

An atomic absorption spectrometer is an instrument which is used to analyse the concentrations of metals in solution. Sixty-eight elements can be determined directly over a wide range of concentrations from ppb to per cent levels, with good precision—typically better than 1 % RSD. Sample preparation is generally simple and frequently involves little more than dissolution in an appropriate acid. The instrument is easy to tune and operate.

Basic Principles of Atomic Absorption

Atomic absorption spectroscopy (AAS) is based upon the principle that free atoms in the ground state can absorb light of a certain wavelength. Absorption for each element is specific, no other elements absorb this wavelength. When a solution having a mixture of metallic species is introduced into the flame, the solvent evaporates and vapour of metallic species is obtained. Some of metal atoms can be raised to an energy level sufficiently high to emit characteristics radiation of metal—a phenomenon that is used in flame photometry. Here a large amount of metal atoms remains in non-emitting ground state.



Instruments of Atomic Absorption Spectroscopy:

The apparatus consists of:

- (1) Radiant Source.
- (2) Atomizer
- (3) Monochromator
- (4) Lenses and Slits and
- (5) Detectors.

Radiant Source:

For atomic absorption spectroscopy the radiation source is a hollow cathode lamp.

It consists of the following parts:

- (i) Cathode: cathode is made of the element to be determined or coated with it.
- (ii) Anode: Anode is made of tungsten, zirconium or nickel.
- (iii) Window is made of Pyrex glass depending on wavelength of emitted radiation.
- (iv) The lamp is filled with neon or argon gas.

Hollow cathode lamps (HCL) are the most common radiation source in AAS. Inside the sealed lamp, filled with argon or neon gas at low pressure, is a cylindrical metal cathode containing the element of interest and an anode. A high voltage is applied across the anode and cathode, resulting in an ionization of the fill gas. The gas ions are accelerated towards the cathode and, upon impact on the cathode, sputter cathode material that is excited in the glow discharge to emit the radiation of the sputtered material, i.e., the element of interest.

(2) Atomizer:

Elements to be analysed need to be in atomic state. Atomization is separation of particles into individual molecules and breaking molecules into atoms. This is done by exposing the analyte to high temperatures in a flame or graphite furnace. Generally, burners are used to break the liquid sample into droplets which are then allowed to enter into flame. The droplets are then evaporated and sample element is left in residue. The residue is then decomposed by flame. Thus, in this process the sample is reduced to atoms.

The role of the atom cell is to primarily dissolve a liquid sample and then the solid particles are vaporized into their free gaseous ground state form. In this form atoms will be available to absorb radiation emitted from the light source and thus generate a measurable signal proportional to concentration.

Atomization steps:

1. **Desolvation (solvent evaporate)**
2. **Volatilization (Gaseous molecules formation)**
3. **Dissociation (Molecules dissociates and produce atomic gas)**
4. **Ionization**
5. **Excitation**

Nebulizer: A device for converting the solution into fine spray or droplets.

(3) Monochromators: Generally, the monochromators are gratings and prisms.

This is a very important part in an AA spectrometer. It is used to separate out all of the thousands of lines. Without a good monochromator, detection limits are severely compromised.

A monochromator is used to select the specific wavelength of light which is absorbed by the sample, and to exclude other wavelengths. The selection of the specific light allows the determination of the selected element in the presence of others.

(4) Filters or slits:

Filters or slits are used for isolation of required spectral line if element has a simple line spectrum.

Detector and Read out Device

The light selected by the monochromator is directed onto a detector that is typically a photomultiplier tube, whose function is to convert the light signal into an electrical signal proportional to the light intensity.

The processing of electrical signal is fulfilled by a signal amplifier. The signal could be displayed for readout, or further fed into a data station for printout by the requested format.

Application:

1. Atomic spectroscopy is used for quantitative analysis of metal elements in water, soil, plant material and ceramics.
2. In health care, it is used to analyze ionic metal elements in blood, saliva, urine samples. The elements analyzed routinely include sodium, potassium, magnesium, calcium and zinc.
3. To determine heavy metals like iron, manganese, copper, zinc, mercury, lead, nickel, and in urine and blood.
4. This analysis is essential in case of heavy metal poisoning. Since heavy metal poisoning is mostly lethal a regular monitoring of poison levels in the patient blood are essential.
5. To determine metal elements like copper, nickel and zinc in the food industry.
6. To estimate Lead in petroleum products.
7. To determine metal concentrations in groundwater and bore well samplings before using for drinking and irrigation.

Nebulizer: A nebulizer is a device that introduces a liquid sample into atomizer. A pneumatic nebulizer converts a sample solution into an aerosol of tiny droplets using a jet of compressed gas.



Sputtering: Argon or neon gas is ionized by an applied voltage on the cell, causing the argon ions to accelerate to the cathode where they interact with the sample of interest and eject atoms. This process is called sputtering, **the ejection of atoms from a sample as a result of bombardment by energetic species.**

GREEN CHEMISTRY

- Green Chemistry is the utilization of a set of principles that reduces or eliminates the use or generation of hazardous substances in the design, manufacture and application of chemical products.
- Green Chemistry is a recent approach to design of energy efficient processes and the best form of waste disposal.
- The awareness among the organic chemists to practice green chemical routes for organic transformations is significantly increasing in the place of mineral acids, mild solid acids or clays are used. The reactions are carried out in organized media or in green solvents.

Green Chemistry is about:

- Waste Minimization at Source
- Use of Catalysts in place of Reagents
- Using Non-Toxic Reagents
- Use of Renewable Resources
- Improved Atom Efficiency
- Use of Solvent Free or Recyclable Environmentally Benign Solvent systems

12 Principles of Green Chemistry:

1. Prevention. It is better to prevent waste than to treat or clean up waste after it is formed.

2. Atom Economy. Synthetic methods should be designed to maximize the incorporation of all materials used in the process into the final product.

3. Less Hazardous Chemical Synthesis. Whenever practicable, synthetic methodologies should be designed to use and generate substances that possess little or no toxicity to human health and the environment.

4. Designing Safer Chemicals. Chemical products should be designed to preserve efficacy of the function while reducing toxicity.

5. Safer Solvents and Auxiliaries. The use of auxiliary substances (solvents, separation agents, etc.) should be made unnecessary whenever possible and, when used, innocuous(harmless).

6. Design for Energy Efficiency. Energy requirements should be recognized for their environmental and economic impacts and should be minimized. Synthetic methods should be conducted at ambient temperature and pressure.

7. Use of Renewable Feedstocks. A raw material or feedstock should be renewable rather than depleting whenever technically and economically practical. 6 Green Chemistry-Ramesh - GEC Kozhikode

8.Reduce Derivatives. Unnecessary derivatization (blocking group, protection/deprotection, temporary modification of physical/chemical processes) should be avoided whenever possible.

9. Catalysis. Catalytic reagents (as selective as possible) are superior to stoichiometric reagents.

10. Design for Degradation. Chemical products should be designed so that at the end of their function they do not persist in the environment and instead break down into innocuous degradation products.

11. Real-time Analysis for Pollution Prevention. Analytical methodologies need to be further developed to allow for real-time in-process monitoring and control prior to the formation of hazardous substances.

12. Inherently Safer Chemistry for Accident Prevention. Substance and the form of a substance used in a chemical process should be chosen so as to minimize the potential for chemical accidents, including releases, explosions, and fires.

Examples of green chemistry

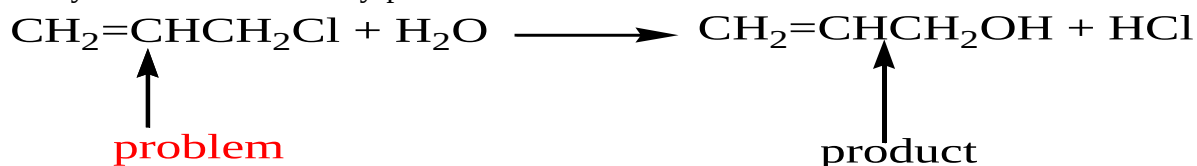
- If the chemical reaction of the type
$$A + B \longrightarrow P + W$$
- Find alternate A or B to avoid W

Disinfection of water:

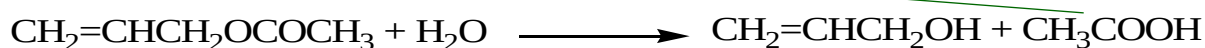
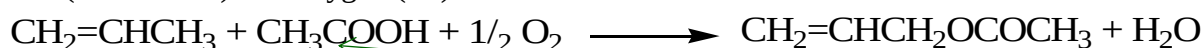
- Disinfection of water by chlorination. Chlorine oxidizes the pathogens there by killing them, but at the same time forms harmful chlorinated compounds.
- A remedy is to use another oxidant, such as O₃ or supercritical water oxidation.

Production of allyl alcohol:

- **Traditional route:** Alkaline hydrolysis of allyl chloride, which generates the product and hydrochloric acid as a by-product



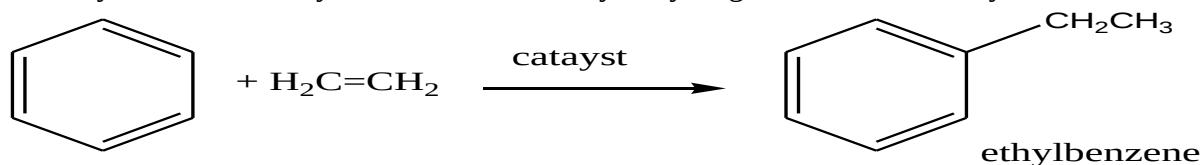
- **Greener route,** to avoid chlorine: Two-step using propylene (CH₂=CHCH₃), acetic acid (CH₃COOH) and oxygen (O₂)



- **Added benefit:** The acetic acid produced in the 2nd reaction can be recovered and used again for the 1st reaction, leaving no unwanted by-product.

Production of Styrene:

- **Traditional route:** Two-step method starting with benzene, (which is carcinogenic) and ethylene to form ethylbenzene, followed by dehydrogenation to obtain styrene



- **Greener route:** To avoid benzene, start with xylene (cheapest source of aromatics and environmentally safer than benzene).

